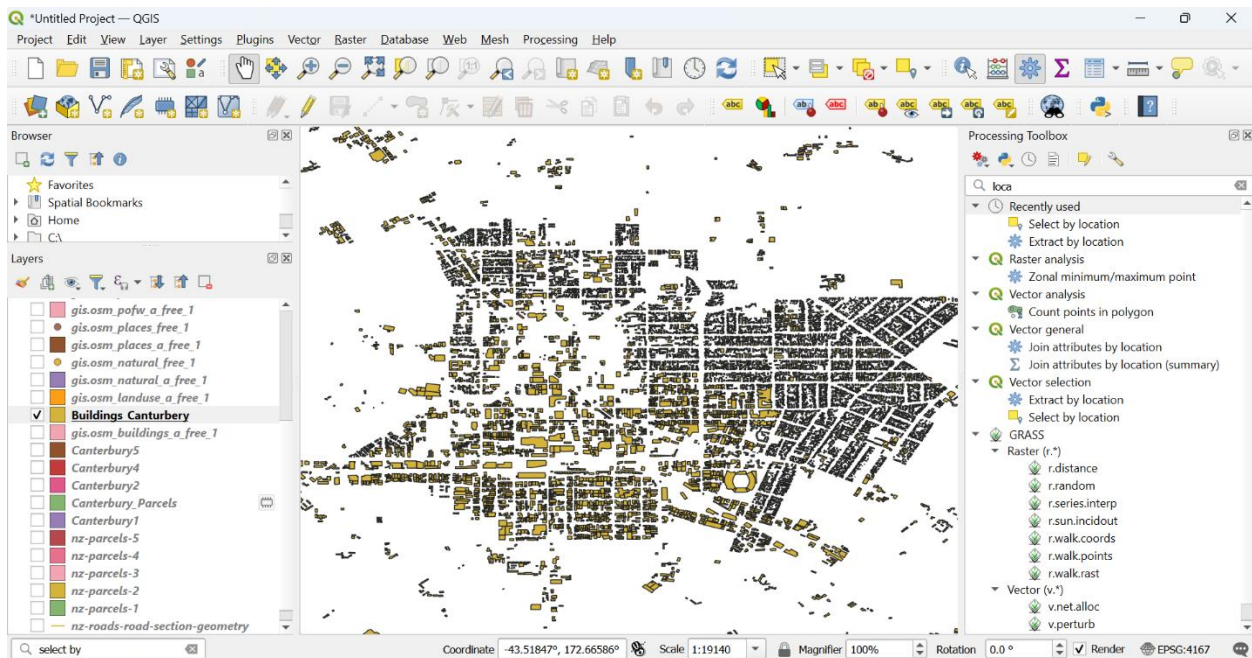


At-Home Activity 3: Querying with Geographic Data

Started a new project in QGIS and added layers of Open Street Map NZ, NZ parcels, roads, place names (Delimited Text Layer). Using attribute and spatial queries created the desired layers. Process and screenshots are below:

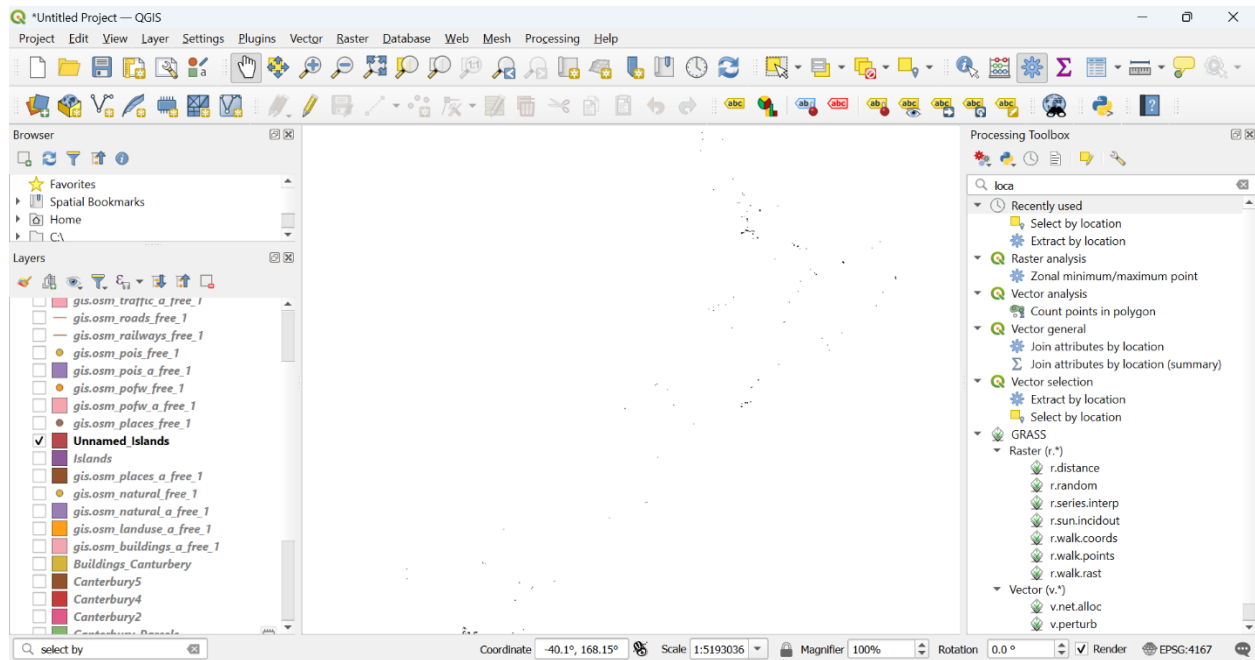
1. Buildings in Canterbury:

From several parcel layers selected ‘Canterbury’ using attribute query View Attribute Table> Select Features Using Expression> “land_distr” = ‘Canterbury’ and saved the layers. Then Merger the vector layers by the option Processing> Toolbox> Vector General > Merge Vector Layers named it “Canterbury_Parcels”. Then using spatial query Processing> Toolbox> Select by Location> buildings layer> intersect> Canterbury_Parcels> Run. Then Saved the layer naming “Buildings_Canterbury”. This new layer shows all the buildings in Canterbury. Here is the screenshot of the zoomed map view.

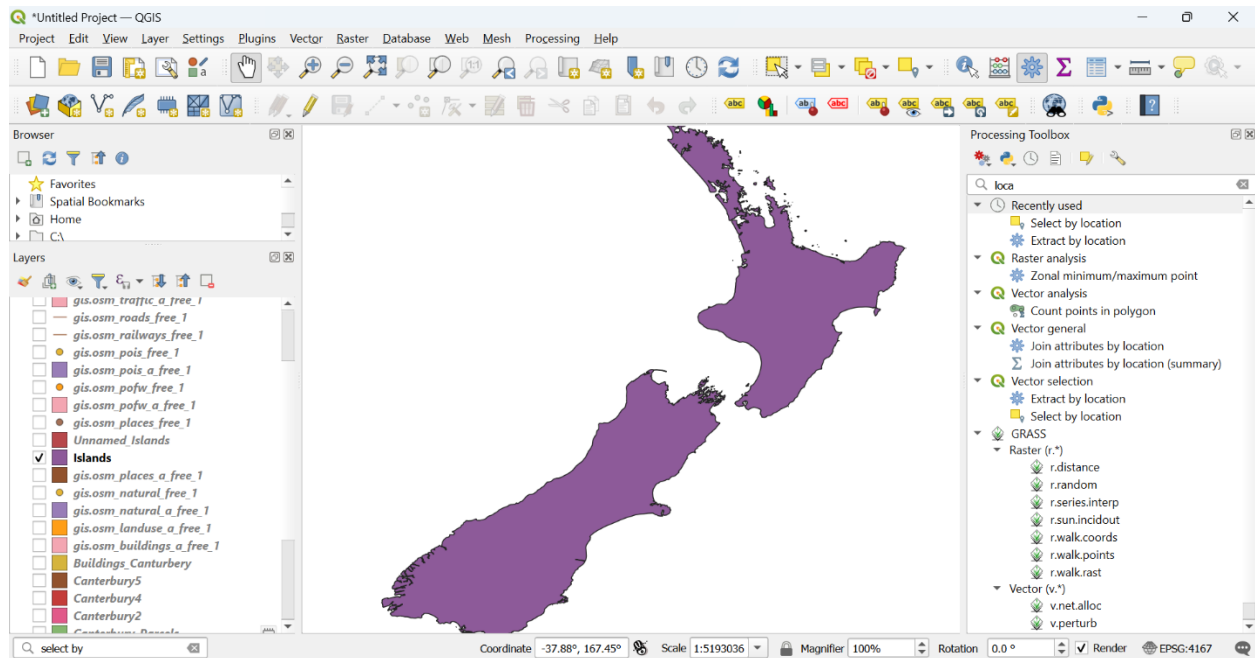


2. Unnamed Islands:

From places layer, selected all the islands using attribute query “fclass = ‘Island’” and saved as a new layer named “Islands”. Then using another attribute query “name is null” on the “Islands” layer and saved selection as a layer called “Unnamed Islands”. This is how got the unnamed islands. Here is the view of the map:



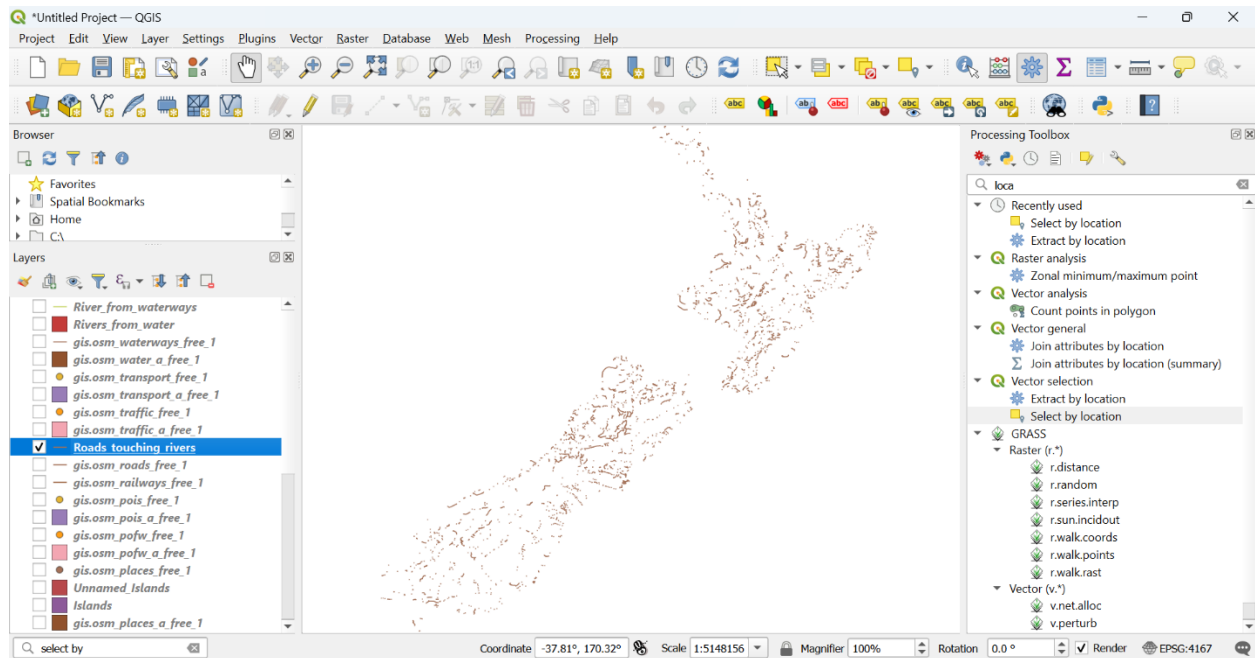
And noticed one strange thing about the dataset. When viewing the Islands layer that includes all the islands- it shows whole NZ map structure. It indicates the two big part of NZ, one is north island and another is south island along with all the small islands around these. Here is the view:



3. Roads that touch rivers:

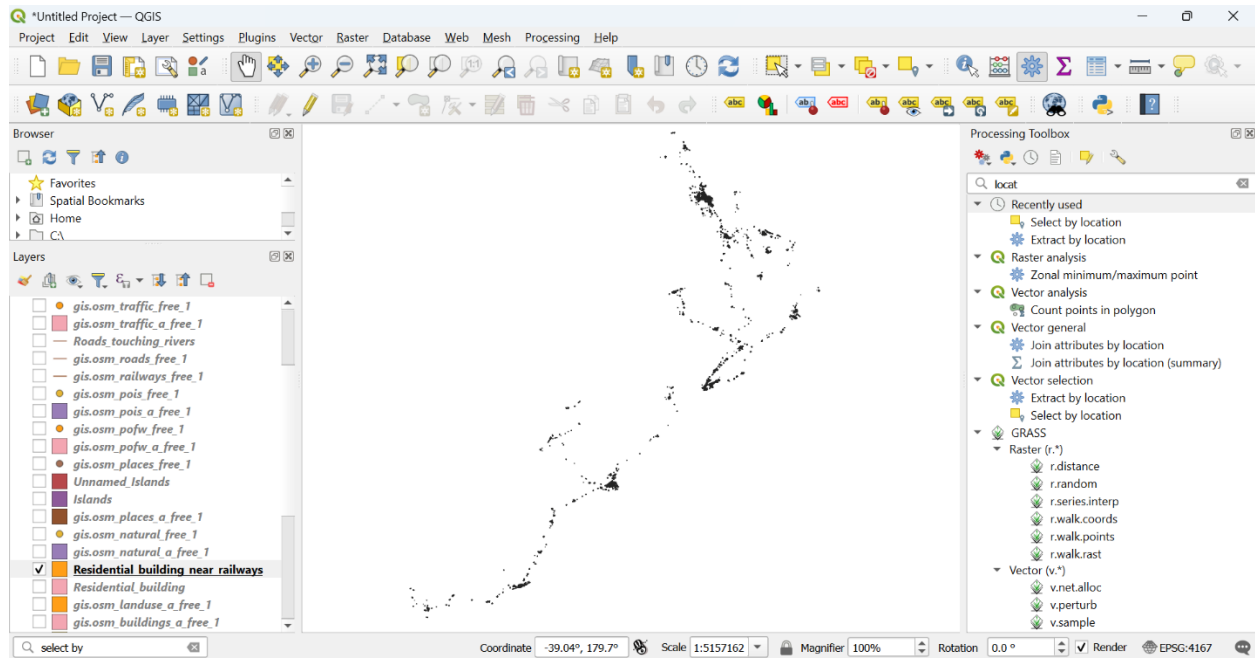
This time, selected rivers from waterways layer using attribute query “fclass = ‘river’” and saved as a new layer named “Rivers_from_waterways”. Then using spatial query selected geometric predicate ‘touch’ between the rivers and roads layers and saved a new layer named

“roads_touching_rivers”. The new layer illustrates all the roads that touch rivers. Here is the view:



4. Residential buildings near railways:

To find out the desired output, selected residential buildings using the attribute query from layer landuse using “fclass = residential” query and saved selection as “Residential_building”. As there was no “near” or “within range” geometric predicate, had to find out another way. Using Vector Geometry> Buffer created a buffer layer with Distance value = 0.1 and segment value 0.1. Then run spatial query on buffered layer and “Residential_building” using geometric predicate intersect to find out the residential buildings near railways. Saved selected features as a new layer called “Residential_buildings_near_railways”. Here is the view:



For Attribute Query process used:

This allows to apply an expression to filter specific features within the layer based on attribute values. It also narrows the big dataset to work with.

Right Click on Layer> View Attribute Table> Select Features Using Expression> ****expression****> Select Features- then Save Selected Features As.

For Spatial Query process used:

This process allows to select features based on their spatial relationship to another layer.

Processing> Toolbox> Select by Location> Layer 1 > Geometric Predicate> Layer 2> Run- then Save Selected Features As.